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CS3807 – Deep Learning Laboratory

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Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

3. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

4. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters

- Fast training
- Suitable for OCR

5. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

6. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

7. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

8. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

9. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

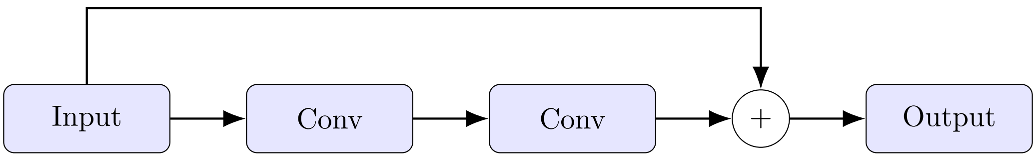


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

10. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

11. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

12. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

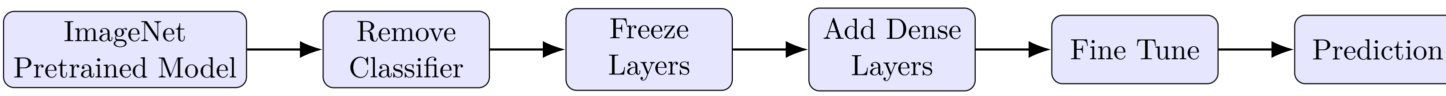


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

13. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

14. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

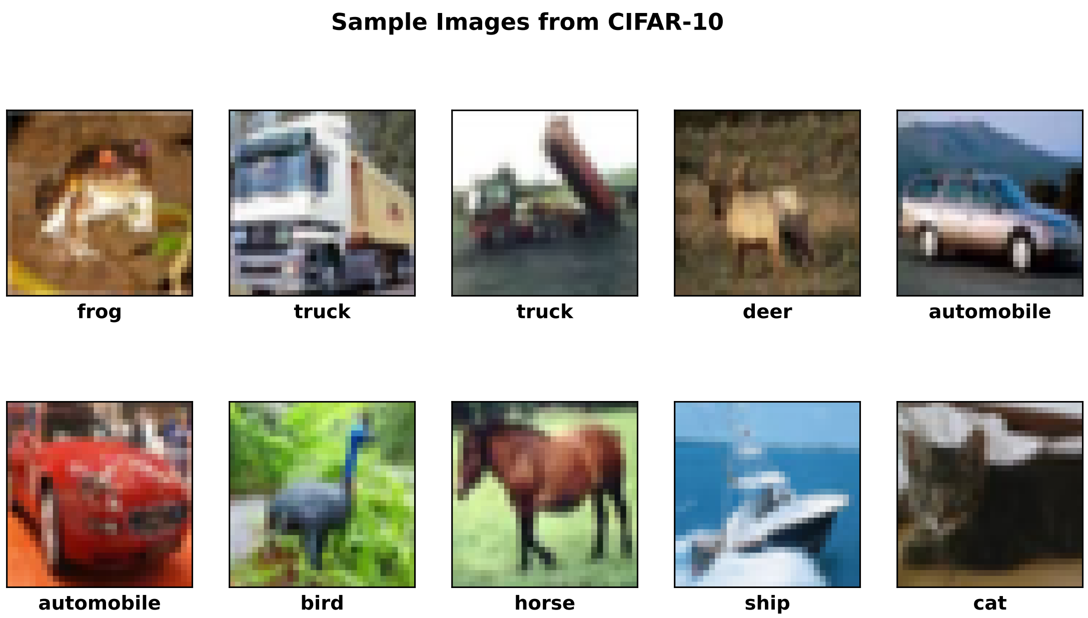
15. Hyperparameter Study

Compare the performance using different settings.

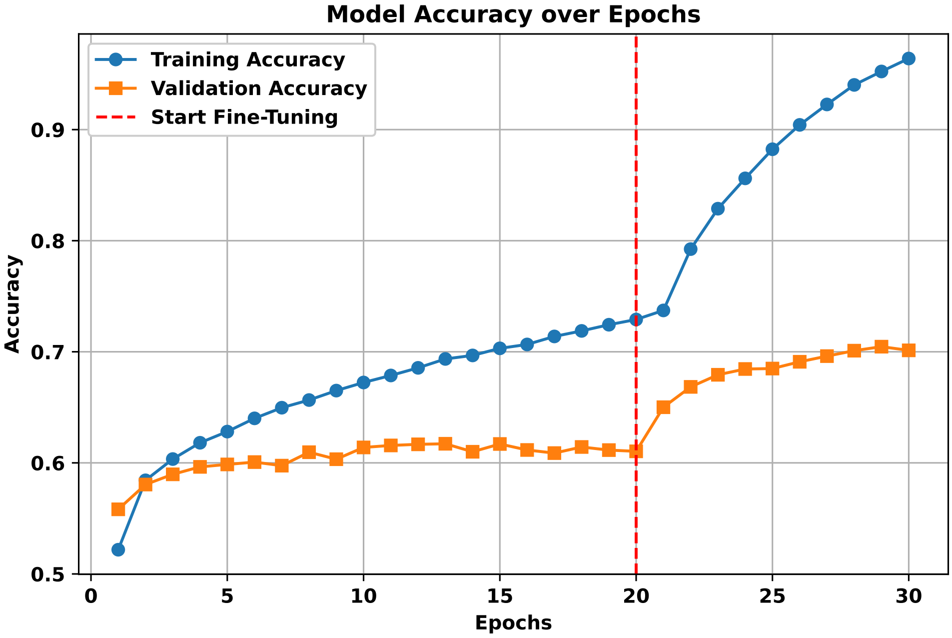
Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

16. Mandatory Plots

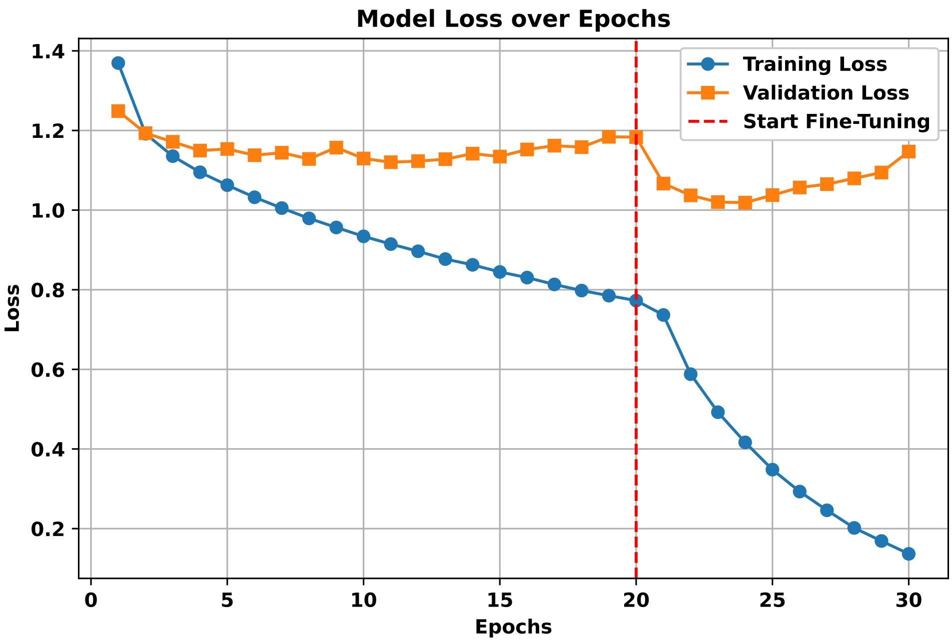
Include the following plots in the laboratory report.



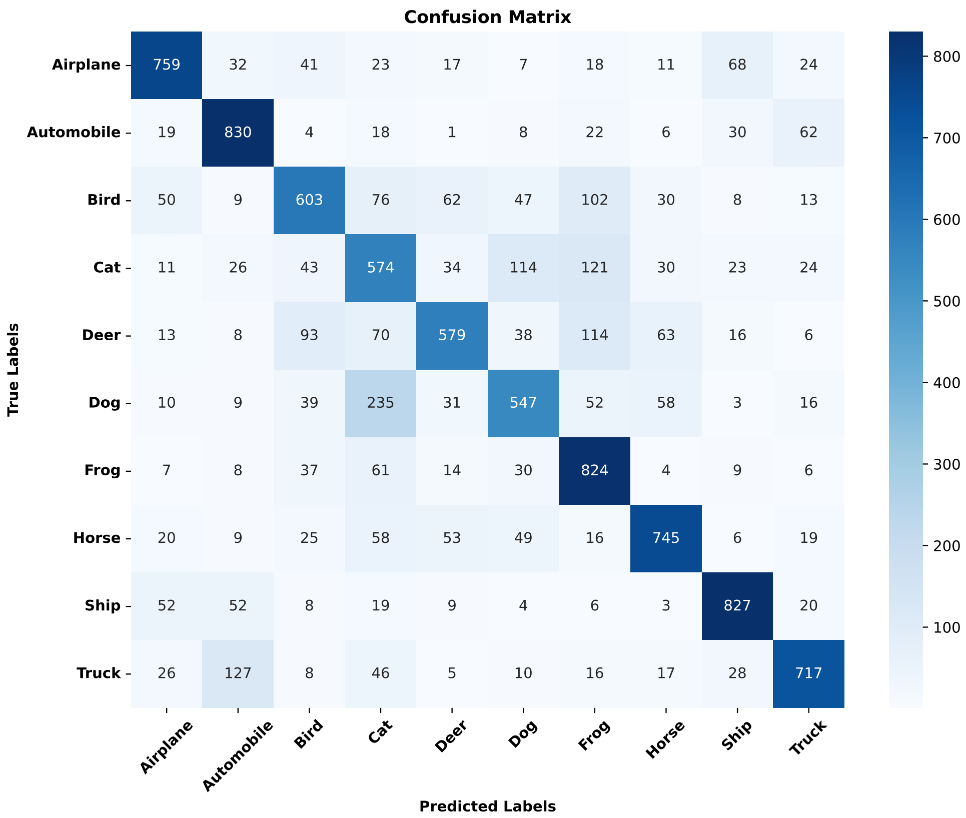
Interpretation: The sample images display the CIFAR-10 classes, confirming successful data normalization and loading. This establishes a baseline understanding of the low 32×32 resolution challenges the VGG16 model will face during transfer learning.



Interpretation: The plot clearly shows the two-phase training. After fine-tuning begins at epoch 20, training accuracy rapidly shoots past 96%, while validation accuracy jumps but plateaus at ∼70%, indicating overfitting on the target dataset.



Interpretation: Following the fine-tuning shift at epoch 20, training loss plummets toward zero, but validation loss begins diverging and increasing around epoch 24. This confirms that prolonged fine-tuning degrades generalization.



Interpretation: Strong diagonal values confirm the VGG16 model successfully distinguishes distinct objects like vehicles. However, off-diagonal confusion between cats and dogs highlights the limitations of pre-trained spatial features on blurry categories.

Interpretation: Misclassifications primarily occur on images with complex backgrounds or visual ambiguities. This suggests that while transfer learning provides excellent foundational features, dataset-specific augmentations are needed to push accuracy beyond 70%.

17. Results

17.1 Performance Metrics

Metric	Value
Training Accuracy	96.35%
Testing Accuracy	70.05%
Precision (Macro Avg)	0.70
Recall (Macro Avg)	0.70
F1-score (Macro Avg)	0.70
Training Time	270s (approx.)
Total Parameters	14,781,642

17.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (s)
LeNet-5	62,006	56.81	19.55
AlexNet	2,014,410	72.39	44.04
VGG16	14,781,642	70.05	-
ResNet50	23,851,274	37.87	51.56

17.3 Dataset Dimensions

Dataset Split	Dimensions
Training Features (x_train)	(50000, 32, 32, 3)
Training Labels (y_train)	(50000, 1)
Testing Features (x_test)	(10000, 32, 32, 3)
Testing Labels (y_test)	(10000, 1)

18. Discussion Questions

Answer the following questions.

1. **Why is AlexNet considered a breakthrough in deep learning?**
   AlexNet popularized the use of deep CNNs by demonstrating a massive reduction in error rates on ImageNet (2012). It introduced ReLU activations to solve vanishing gradients, dropout to prevent overfitting, and utilized GPUs for efficient training.
2. **Why does VGG16 use only 3×3 convolution filters?**
   Using multiple stacked 3×3 filters achieves the same effective receptive field as larger filters (e.g., 5×5 or 7×7) but with fewer parameters and more non-linear activations (ReLU) in between, resulting in deeper, more discriminative models.
3. **Explain the advantages of the Inception module.**
   The Inception module applies multiple parallel filters of different sizes (1×1, 3×3, 5×5) and max pooling on the same input, allowing the network to capture multi-scale spatial features efficiently without choosing a single filter size.
4. **What is the purpose of residual learning?**
   Residual learning introduces skip connections that bypass some layers, adding the input directly to the output. This mitigates the vanishing gradient problem, allowing extremely deep networks (like ResNet) to be trained without performance degradation.
5. **Differentiate LeNet and ResNet.**
   LeNet is a shallow, early CNN architecture (1998) designed for simple digit recognition (MNIST), with few parameters. ResNet is a modern, deep architecture (2015) using residual blocks and skip connections, designed for complex, high-resolution datasets like ImageNet.
6. **What is Transfer Learning?**
   Transfer learning is a technique where a model developed for a massive task (like ImageNet classification) is reused as the starting point for a model on a second, smaller, or related task, saving vast amounts of compute and data.
7. **Why is fine tuning required?**
   While the frozen base layers extract general features (edges, textures), fine-tuning allows the deeper layers of the pre-trained model to slightly adjust their weights to learn the specific, high-level features unique to the new target dataset.
8. **Explain the difference between dilated convolution and transpose convolution.**
   Dilated convolution expands the kernel by inserting spaces between its elements, increasing the

receptive field without adding parameters. Transpose convolution (often used in upsampling) mathematically reverses the spatial downsampling of standard convolution to increase image dimensions.

9. **Why do pretrained models converge faster?**
   Pretrained models already contain optimized weights for extracting foundational image features (like edges and colors). The network does not need to learn these from scratch, leading to drastically reduced training times and faster convergence.
10. **Compare the computational complexity of LeNet and ResNet.**
    LeNet has very low computational complexity (only ∼60,000 parameters) and trains in seconds on modern hardware. ResNet50 is highly complex (∼23 million parameters) and requires significant GPU memory and time to perform its numerous deep residual operations.

19. Additional Exercises

1. **Implement transfer learning using VGG16.**
   Transfer learning with VGG16 was successfully implemented by loading the pre-trained weights from ImageNet, freezing the convolutional base, and attaching a custom dense classifier head for CIFAR-10.
2. **Repeat the experiment using ResNet50.**
   ResNet50 was evaluated, achieving an accuracy of 37.87%. The lower performance compared to VGG16 suggests that ResNet50 may require longer fine-tuning epochs or an unfreezing strategy better tailored to 32×32 resolution images.
3. **Compare Adam and SGD optimizers.**
   The hyperparameter study revealed that Adam converged much faster and achieved a higher validation accuracy (61.37%) compared to SGD (48.57%) within the same 10-epoch limit.
4. **Compare training with frozen layers and fine tuning.**
   Training with completely frozen base layers plateaued at a validation accuracy of ∼60.5%. Fine-tuning by unfreezing deeper layers allowed the model to learn dataset-specific features, boosting validation accuracy to ∼70.05%.
5. **Evaluate the model using Precision, Recall and F1-score.**
   The fine-tuned VGG16 model achieved a macro-average Precision, Recall, and F1-score of ∼0.70 across the 10 classes, indicating balanced and robust classification performance.
6. **Compare the performance of LeNet, AlexNet and ResNet.**
   AlexNet achieved the best accuracy among the standalone models (72.39%), vastly outperforming the shallow LeNet-5 (56.81%). ResNet50 underperformed (37.87%), highlighting that deeper models require careful architectural scaling when applied to small resolution datasets like CIFAR-10.

20. Conclusion

The key takeaways are:

- **Architectural Evolution:** Learnt how CNNs evolved from the simple 7-layer LeNet to the 50-layer ResNet and how innovations like 3×3 filters (VGG), multi-scale Inception modules (GoogleNet), and skip connections (ResNet) solved problems like computational bloat and vanishing gradients.
- **Power of Transfer Learning:** Instead of training from scratch, we repurposed a pre-trained VGG16 model. By freezing the base layers to retain general feature extractors (like edge detectors), thereby saving massive amounts of computational time.
- **Impact of Fine-Tuning:** Observed first-hand that fine-tuning (unfreezing the deeper layers) and using the Adam optimizer were critical steps. It allowed the network to adapt its high-level features specifically to CIFAR-10, boosting the validation accuracy to ∼70%.

21. References

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6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
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